Introduction to Artificial Intelligence

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Assignment: Ethical aspects of AI

In this lab session, you will discuss ethical aspects of AI. You will be working in your teams of 4 students each. You will prepare and present your results during the lab session (no homework – the homework for Assignment 07 continues).

Select at least one of the following groups of questions in your team, discuss those and related questions, perform internet research and summarize your results in the online presentation (ca. 5 minutes).

1. **AI dystopia**: Is the vision of hyper-intelligent machines that soon conquer the world just bold science fiction? Or are there indeed fundamental differences between humans and machines? Why are AI hypes recurring? Is it an age-old fascination / fear of creating a homunculus? Is there financial interest in the hype?
2. **Ubiquitous AI**: In which way change AI applications our daily life? What are benefits and risks? Is there a need for legal frameworks?
3. **Cyborgs**: What are benefits and risks of technologically enhancing human bodies? Is there a need for legal frameworks?
4. **AI and labour market**: Will AI applications continue to be most useful in many domains and business sectors? Will increasing automation continuously decrease the need for human labour? Which forms of labour and sharing of wealth is appropriate in future? Is an unconditional basic income a viable approach?
5. **Legal aspects**: Who should be accountable when an AI system fails? Which regulatory measures are needed? Which discussions are needed in society?
6. **Responsibility**: Which kinds of decisions may be left to AI systems; which ones must be taken by humans? Which regulatory measures are needed? Which discussions are needed in society?
7. **Bias in ML**: Name prominent examples of bias in ML. What are the causes? What are the effects? How to prevent / reduce bias in ML?
8. **Vulnerable AI**: What are techniques for attacking AI systems? What are the consequences? What are measures to prevent this?
9. **Green AI**: What are positive and negative impacts of AI on environmental issues like climate change? What are measures to increase positive impact? What are measures to decrease negative impact?
10. **Other aspects**: What are other impacts of AI on society (positive and negative)? How should they be dealt with?